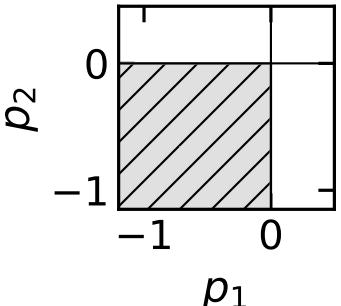
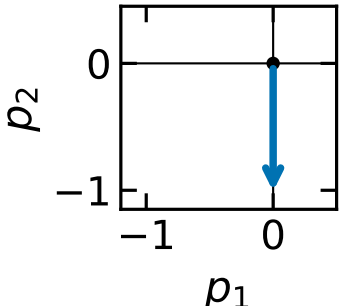
C_1: first-order feasibility



C_2: also zero objective slope